

PROTEIN AND AMINO ACID METABOLISM

MEDICAL AND BIOLOGICAL IMPORTANCE:

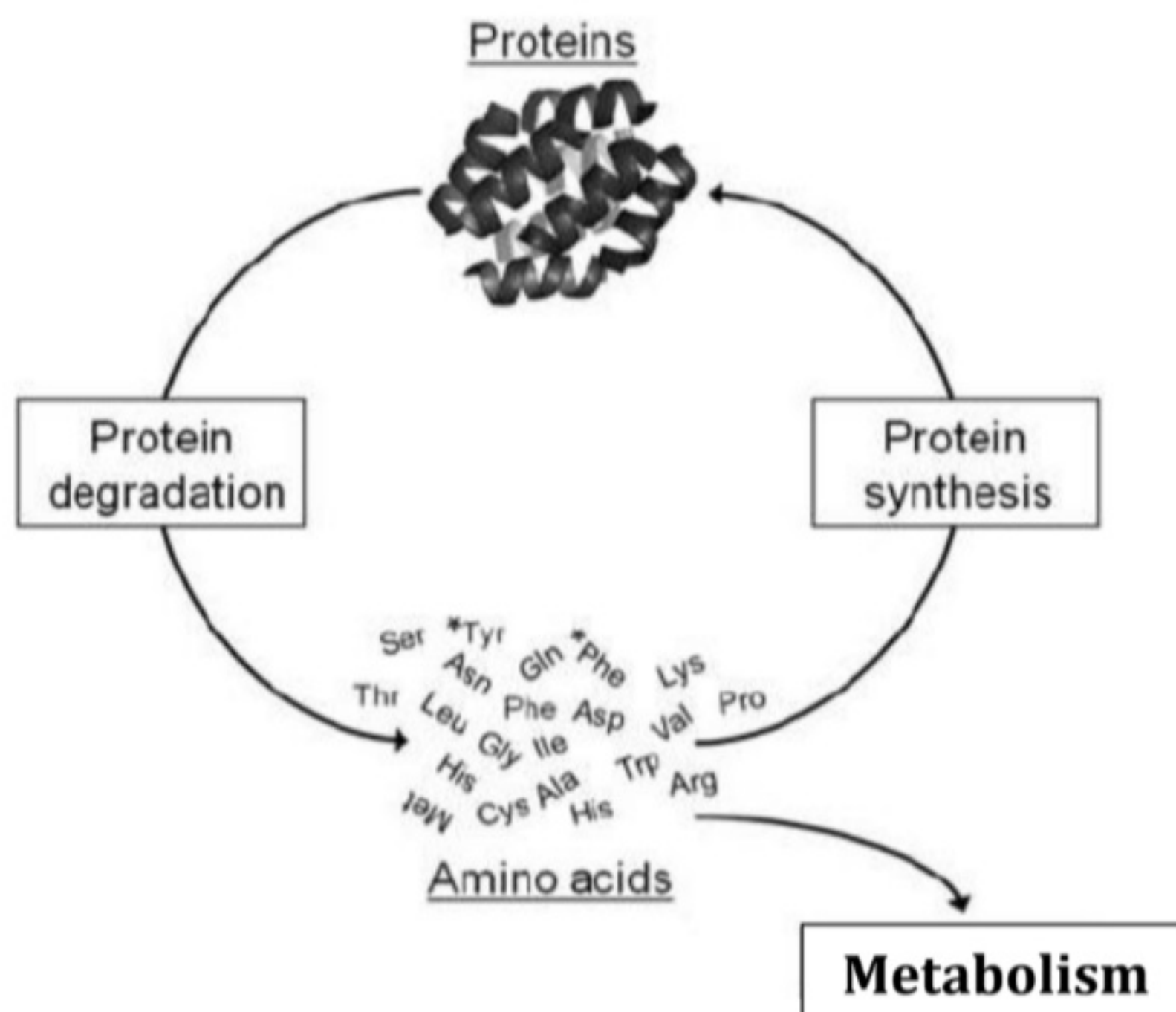
- A 70 kg human adult body contains about 12 kg of protein.
- Body proteins have life times. They undergo degradation and re-synthesis. About 400 gm of body protein is synthesized and degraded per day i.e., about 6 gm of protein is synthesized and broken down per kg body weight per day.
- Aged proteins, damaged or modified proteins and non-functional proteins of the body undergo degradation. Further degradation is one way of controlling enzyme activity. Hence, continuous re-synthesis and degradation of proteins is a quality control mechanism.
- Protein degradation may play important role in shaping tissues and organs during pregnancy and development.
- In starvation, diabetes and tissue injury, protein degradation is more.
- Protein synthesis and degradation is an integral part of cellular adaptation to changed environment.
- Plasma free amino acid concentration ranges from 40 to 60 mg%. Excess amino acids cannot be stored in the body. First amino group is extracted as ammonia and then carbon skeleton is oxidized to produce energy. In starvation carbon skeletons are used for glucose formation. Carbon skeletons of some amino acids produce acetyl-CoA as end product.
- Ammonia, which is toxic to cells is converted to urea in the liver. Conversion of ammonia to urea is impaired in some inherited diseases and liver disease.
- Amino acids are needed for the formation of specialized products like hormones, purines, pyrimidines, porphyrins, vitamins, amines, creatine and glutathione.
- Amino acid degradation is impaired in several inherited diseases due to lack of enzymes.
- Amino acid degradation is more in starvation, diabetes and high protein diet.
- Some cancer cells have high amino acid (asparagine) requirement.

PROTEIN TURN OVER:

- In all forms of life, proteins once formed may not remain forever. Like intermediates of metabolic path ways, proteins are synthesized and degraded. Hence, body protein is in dynamic state. Continuous synthesis and degradation of protein is called as **protein turnover**.
- It is estimated that about **300-400 g** of protein per day is constantly degraded and synthesized which represents body protein turnover.
- The rates of protein synthesis and degradation vary according to physiological needs. The rate of protein synthesis is high during growth, lactation and post operative recovery.
- In starvation, cancer, fever and during morphogenesis rate of degradation of protein is more.

CONTROL OF PROTEIN TURNOVER:

The turnover of the protein is influenced by many factors. A small polypeptide called **ubiquitin** (mol. wt. 8,500) tags with the proteins and facilitates degradation. Certain proteins with amino acid sequence proline, glutamine (one letter code E), serine and threonine (PEST sequence) are rapidly degraded.



NITROGEN BALANCE:

Catabolism of amino acids leads to a net loss of nitrogen from the body. This loss must be compensated by the diet in order to maintain a constant amount of body protein. Nitrogen balance studies evaluate the relationship between the nitrogen intake (in the form of protein) and nitrogen excretion.

Three situations of nitrogen balance are possible as follows:

- 1) Nitrogen equilibrium
- 2) Positive nitrogen balance
- 3) Negative nitrogen balance.

1) NITROGEN EQUILIBRIUM:

- In normal adults, nitrogen intake is equal to nitrogen excretion. The subject is said to be in nitrogen equilibrium or balance.
- In this situation, the rate of body protein synthesis is equal to the rate of degradation.

2) POSITIVE NITROGEN BALANCE:

- In this situation, nitrogen intake > nitrogen excretion, i.e. intake of nitrogen is more than excretion.
- It shows that nitrogen is retained in the body, which means that protein is laid down.
- This occurs in growing **infants** and **pregnant women**.

3) NEGATIVE NITROGEN BALANCE:

- In this situation, nitrogen intake < nitrogen excretion, i.e. nitrogen output exceeds input and this occurs during serious illness and major injury and trauma, in advanced cancer and following failure to ingest adequate or sufficient high quality protein, e.g. in **kwashiorkor** and **marasmus**.
- If the situation is prolonged, it will ultimately lead to death.

CATABOLISM OF AMINO ACIDS:

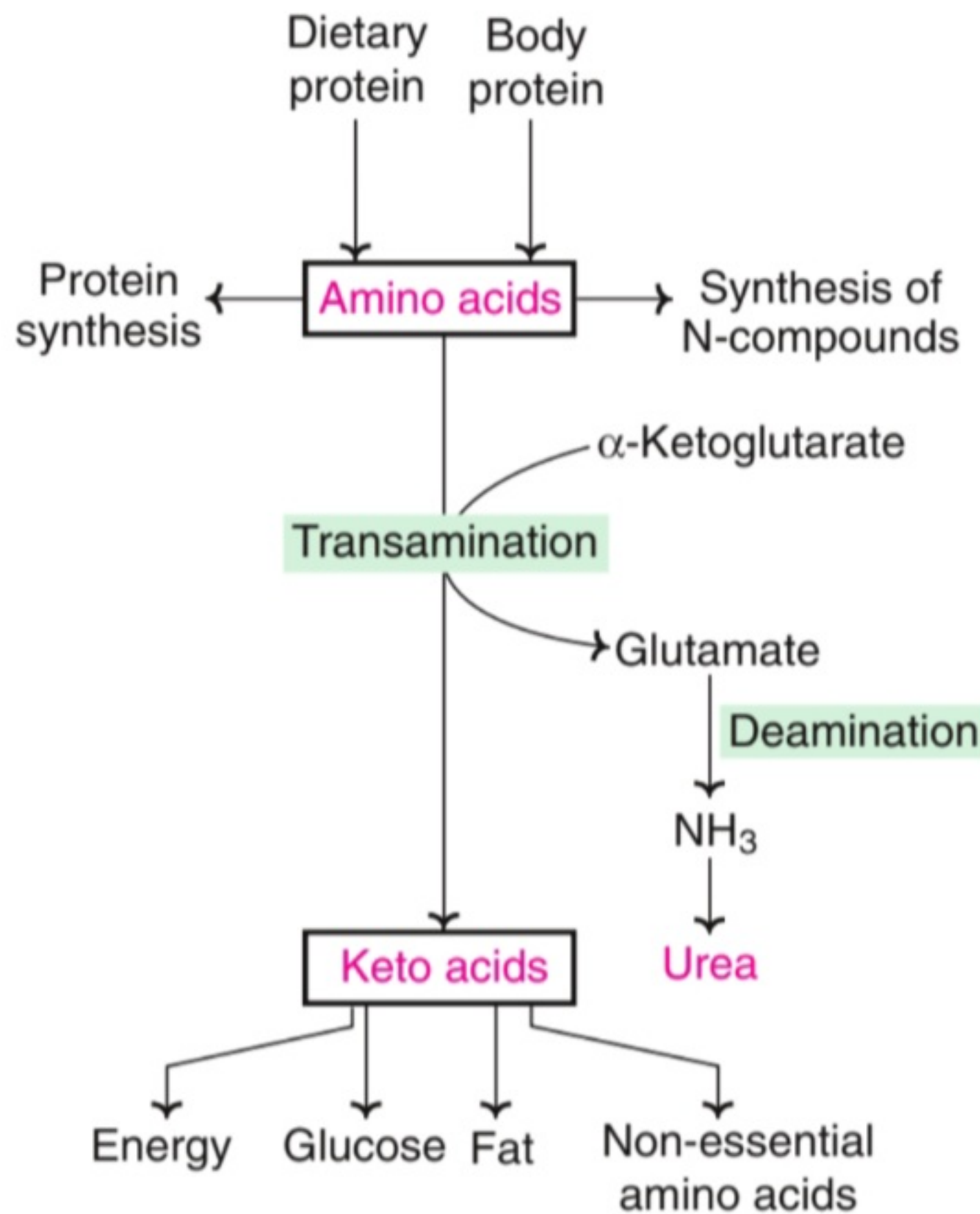


Fig. 15.2 : An overview of amino acid metabolism.

The complete catabolism of amino acids includes following stages:

- 1) The removal of α-amino group in the form of ammonia by following reactions:
 1. **Transamination** by the enzyme aminotransferase also called transaminases.
 2. **Deamination** may be oxidative or nonoxidative
 - a) Oxidative deamination is by glutamate dehydrogenase or amino acid oxidase.
 - b) Nonoxidative deamination is by amino acid dehydratase.
- 2) Disposal of ammonia in the form of urea in the liver by reactions of the **urea cycle**.
- 3) Disposal (catabolism) of the remaining carbon skeleton of amino acid to carbon dioxide and water by reactions of **citric acid cycle**.

TRANSAMINATION:

The transfer of an amino ($-\text{NH}_2$) group from an amino acid to a keto acid is known as **transamination**. This process involves the interconversion of a pair of amino acids and a pair of keto acids, catalysed by a group of enzymes called **transaminases** (recently, aminotransferases).

SALIENT FEATURES OF TRANSAMINATION:

- All transaminases require pyridoxal phosphate (PLP), a coenzyme derived from vitamin B₆.
- Specific transaminases exist for each pair of amino and keto acids. However, only two— namely, aspartate transaminase and alanine transaminase—make a significant contribution for transamination.
- There is no free NH_3 liberated, only the transfer of amino group occurs.
- Transamination is reversible (**Fig.15.3**).
- Transamination is very important for the redistribution of amino groups and **production of non-essential amino acids**, as per the requirement of the cell. It involves both catabolism (degradation) and anabolism (synthesis) of amino acids.
- Transamination diverts the excess amino acids towards **energy generation**.
- The amino acids undergo transamination to finally concentrate nitrogen in glutamate. **Glutamate** is the only amino acid that undergoes oxidative deamination to a significant extent to liberate free NH_3 for urea synthesis.
- All amino acids except lysine, threonine, proline and hydroxyproline participate in transamination.
- Transamination is not restricted to α -amino groups only. For instance, δ -amino group of ornithine is transaminated.
- Serum transaminases are important for diagnostic and prognostic purposes.

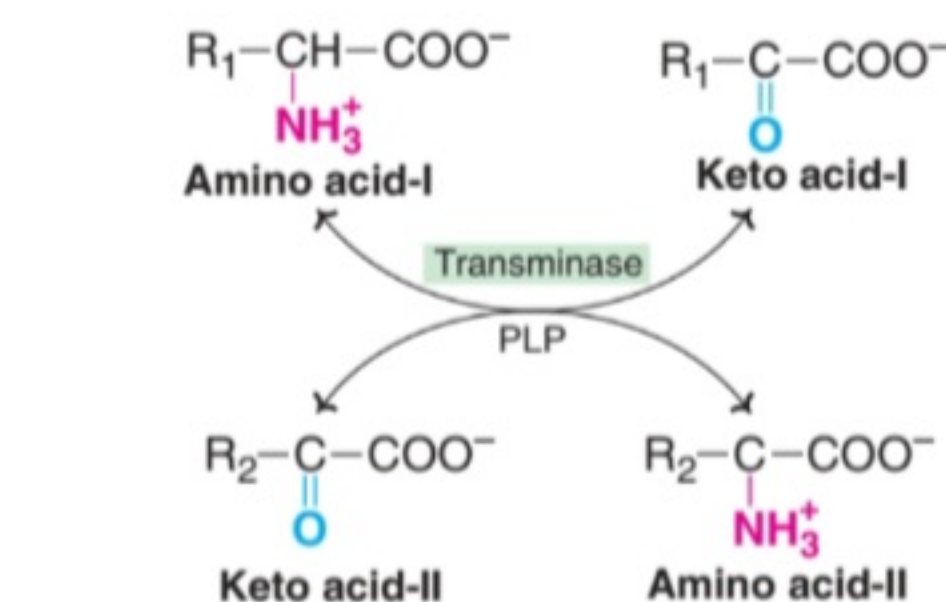
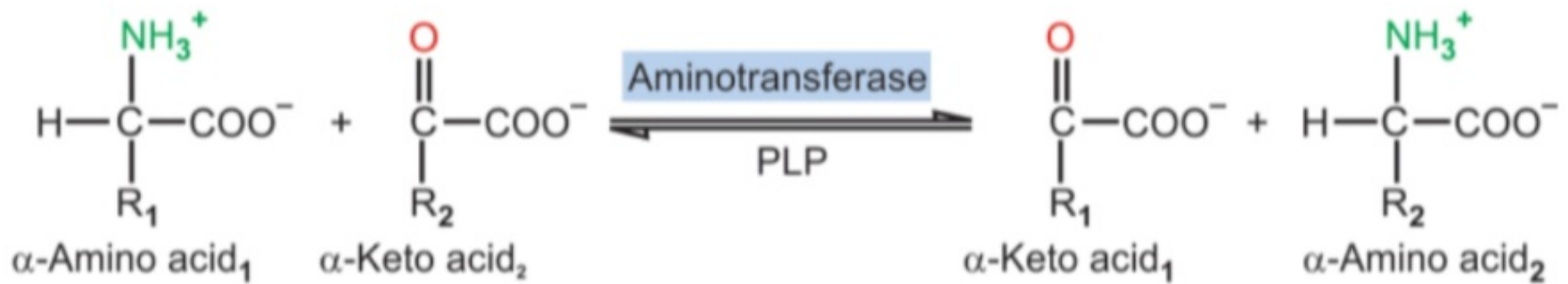


Fig. 15.3 : Transamination reaction.

(or)



MECHANISM OF TRANSAMINATION:

Transamination occurs in two stages (Fig.15.4)

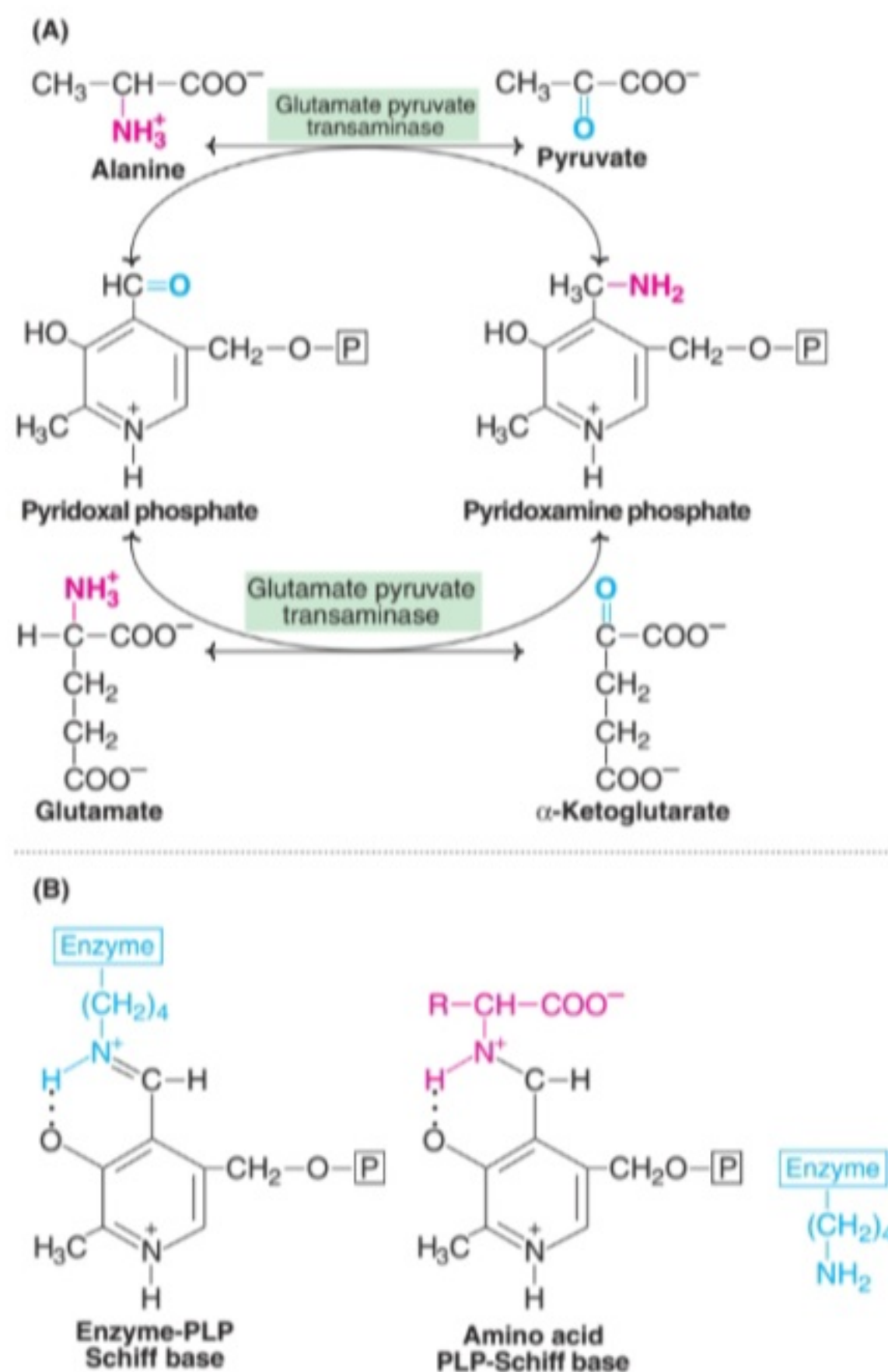


Fig. 15.4 : Mechanism of transamination—(A) Involvement of pyridoxal phosphate (PLP) in the transfer of amino group, (B) Formation of enzyme-PLP-Schiff base and amino acid-PLP-Schiff base.
Note that when the amino acid binds, enzyme separates.

- 1) Transfer of the amino group to the coenzyme pyridoxal phosphate (bound to the coenzyme) to form pyridoxamine phosphate.
- 2) The amino group of pyridoxamine phosphate is then transferred to a keto acid to produce a new amino acid and the enzyme with PLP is regenerated.

All the transaminases require **pyridoxal phosphate** (PLP), a derivative of vitamin B₆. The aldehyde group of PLP is linked with ϵ -amino group of lysine residue, at the active site of the enzyme forming a **Schiff base** (imine linkage). When an amino acid (substrate) comes in contact with the enzyme, it displaces lysine and a new Schiff base linkage is formed. The amino-acid-PLP-Schiff base tightly binds with the enzyme by noncovalent forces. Snell and Braustein proposed a **Ping Pong Bi Bi** mechanism involving a series of intermediates (aldimines and ketimines) in transamination reaction.

DEAMINATION:

The **removal of amino group** from the amino acids as NH₃ is deamination. Transamination (discussed above) involves only the shuffling of amino groups among the amino acids. On the other hand, deamination results in the liberation of ammonia for urea synthesis. Simultaneously, the carbon skeleton of amino acids is converted to keto acids. Deamination may be either oxidative or non-oxidative.

Although transamination and deamination are separately discussed, they occur simultaneously, often involving glutamate as the central molecule. For this reason, some authors use the term **transdeamination** while describing the reactions of transamination and deamination, particularly involving glutamate.

1) OXIDATIVE DEAMINATION:

Oxidative deamination is the **liberation of free ammonia** from the amino group of amino acids **coupled with oxidation**. This takes place mostly in liver and kidney. The purpose of oxidative deamination is to provide NH₃ for urea synthesis and α -keto acids for a variety of reactions, including energy generation.

ROLE OF GLUTAMATE DEHYDROGENASE:

In the process of transamination, the amino groups of most amino acids are transferred to α -ketoglutarate to produce glutamate. Thus, **glutamate** serves as a '**collection centre**' for amino groups in the biological system. Glutamate rapidly undergoes oxidative deamination, catalysed by glutamate dehydrogenase (GDH) to liberate ammonia. This enzyme is unique in that it can utilize either NAD⁺ or NADP⁺ as a coenzyme. Conversion of glutamate to α -keto-glutarate occurs through the formation of an intermediate, α -iminoglutarate (**Fig.15.5**).

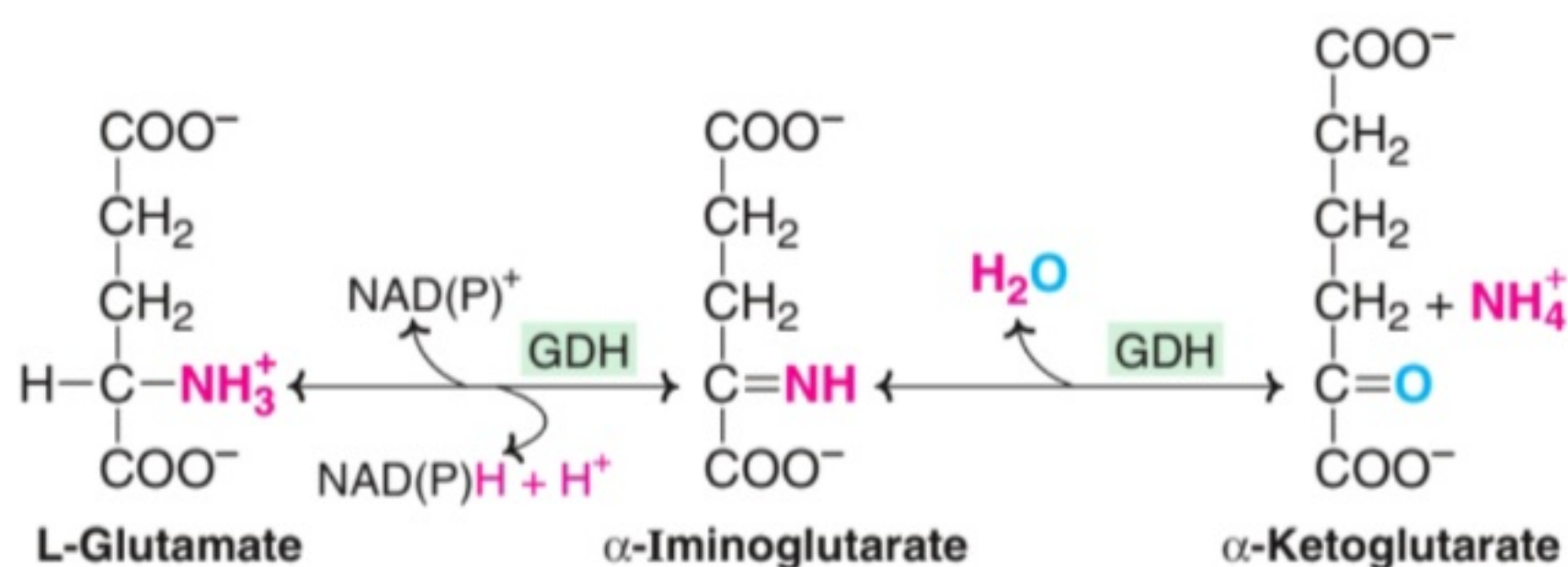


Fig. 15.5 : Oxidation of glutamate by glutamate dehydrogenase (GDH).

Glutamate dehydrogenase catalysed reaction is important as it reversibly links up glutamate metabolism with TCA cycle through α-ketoglutarate. GDH is involved in both catabolic and anabolic reactions.

Regulation of GDH activity: Glutamate dehydrogenase is a zinc containing mitochondrial enzyme. It is a complex enzyme consisting of six identical units with a molecular weight of 56,000 each. GDH is controlled by allosteric regulation. **GTP** and **ATP** inhibit— whereas **GDP** and **ADP activate**—glutamate dehydrogenase. Steroid and thyroid hormones inhibit GDH.

After ingestion of a protein-rich meal, liver glutamate level is elevated. It is converted to α-ketoglutarate with liberation of NH₃. Further, when the cellular energy levels are low, the degradation of glutamate is increased to provide α-ketoglutarate which enters TCA cycle to liberate energy.

OXIDATIVE DEAMINATION BY AMINO ACID OXIDASES:

L-Amino acid oxidase and D-amino acid oxidase are flavoproteins, possessing FMN and FAD, respectively. They act on the corresponding amino acids (L or D) to produce α-keto acids and NH₃. In this reaction, oxygen is reduced to H₂O₂, which is later decomposed by catalase (Fig.15.6).

The activity of **L-amino acid oxidase** is much low while that of **D-amino acid oxidase** is high in tissues (mostly liver and kidney). L-Amino acid oxidase does not act on glycine and dicarboxylic acids. This enzyme, due to its very low activity, does not appear to play any significant role in the amino acid metabolism.

Fate of D-amino acids: D-Amino acids are found in plants and microorganisms. They are, however, not present in the mammalian proteins. But D-amino acids are

regularly taken in the diet and metabolized by the body. D-Amino acid oxidase converts them to the respective D-keto acids by oxidative deamination. The α -keto acids so produced undergo transamination to be converted to L-amino acids which participate in various metabolisms. Keto acids may be oxidized to generate energy or serve as precursors for glucose and fat synthesis. Thus, D-amino acid oxidase is important as it initiates the first step for the **conversion of unnatural D-amino acids to L-amino acids** in the body (Fig.15.7).

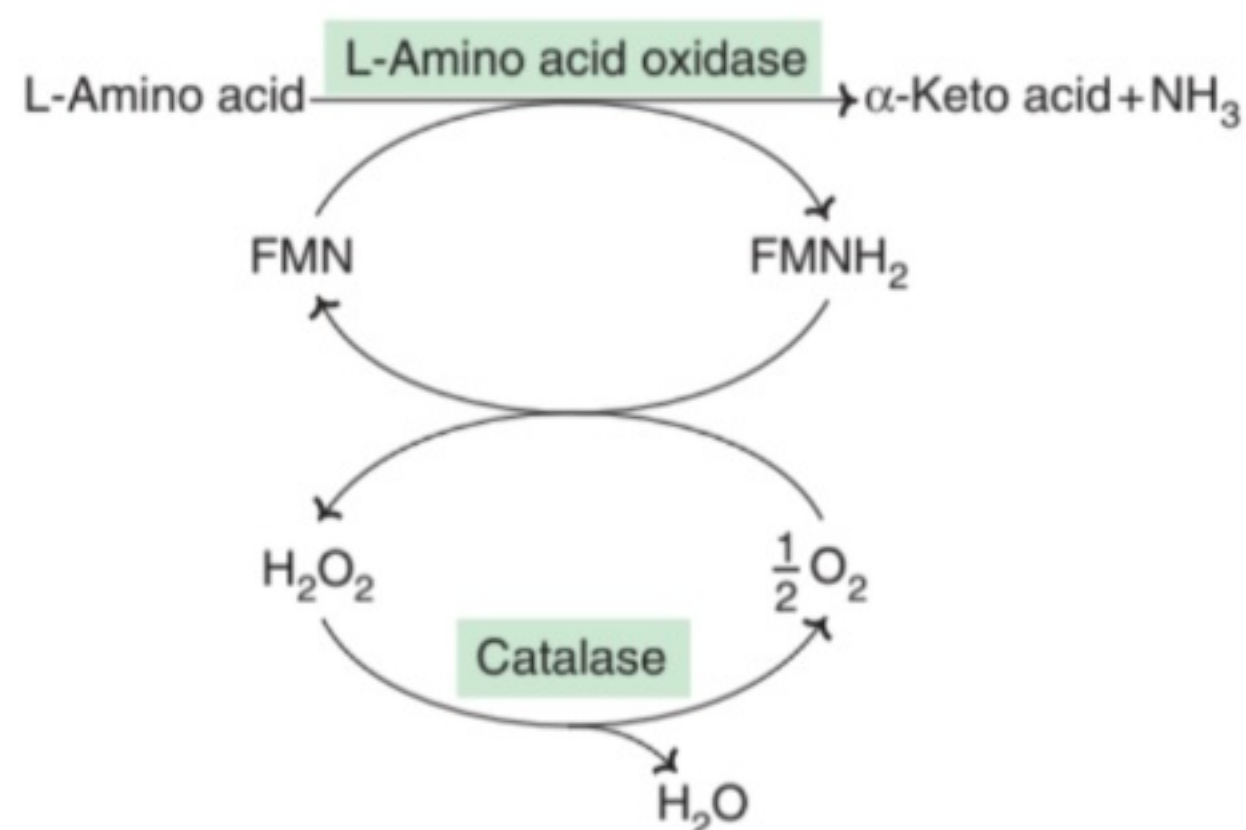


Fig. 15.6 : Oxidative deamination of amino acids.

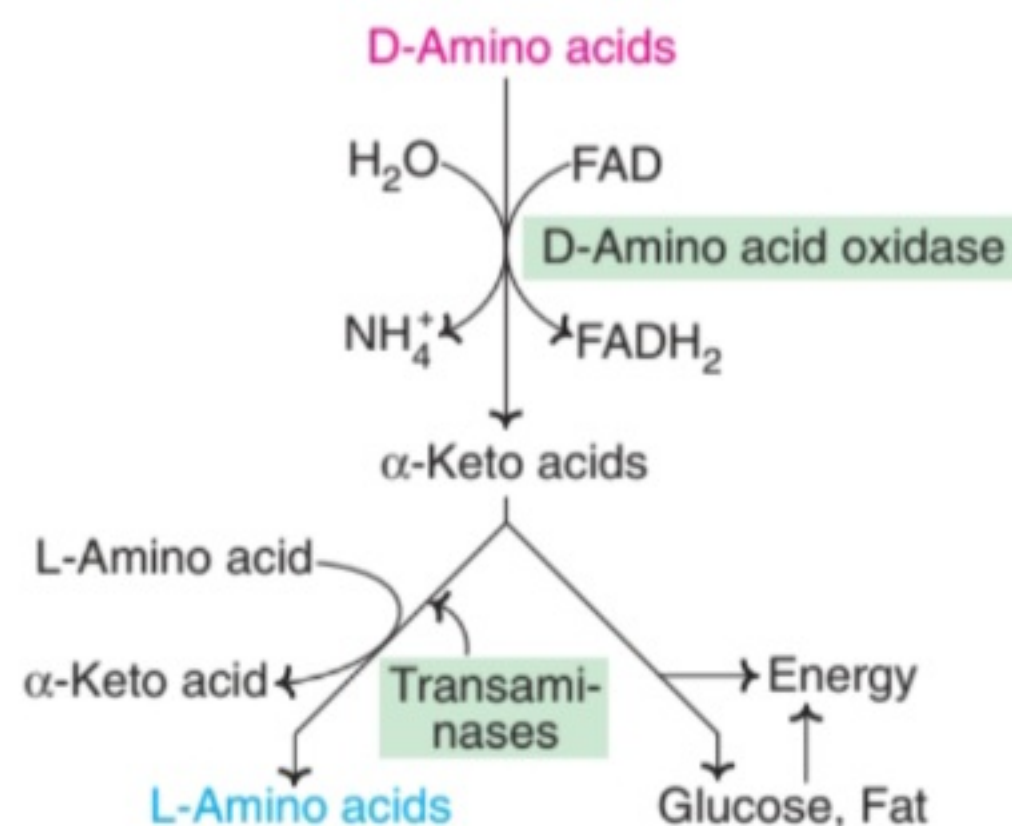
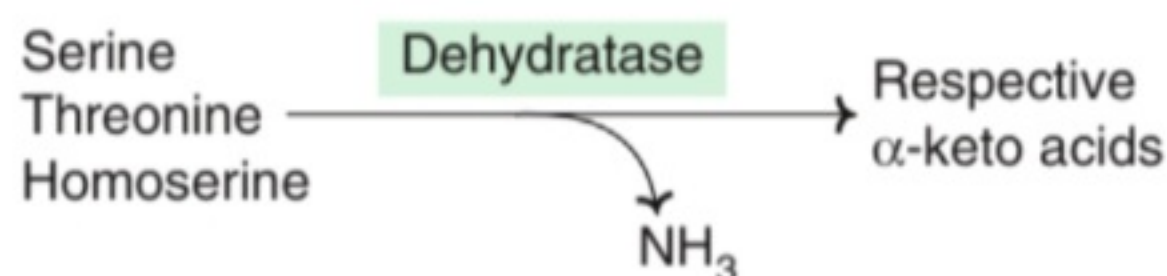


Fig. 15.7 : Metabolic fate of D-amino acids.

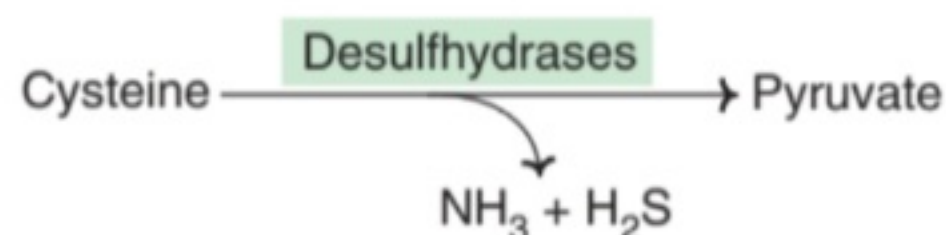
2) NON-OXIDATIVE DEAMINATION:

Some of the amino acids can be deaminated to liberate NH_3 without undergoing oxidation

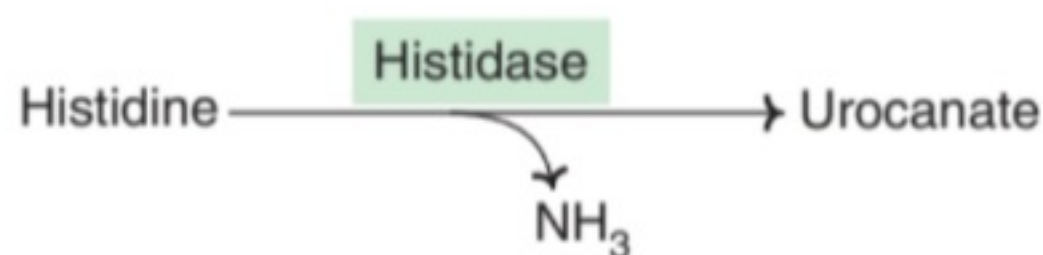
(a) Amino acid dehydrases: Serine, threonine and homoserine are the hydroxy amino acids. They undergo non-oxidative deamination catalysed by PLP-dependent dehydrases (dehydratases).



(b) Amino acid desulfhydrases: The sulfur amino acids, namely cysteine and homocysteine, undergo deamination coupled with desulfhydration to give keto acids.

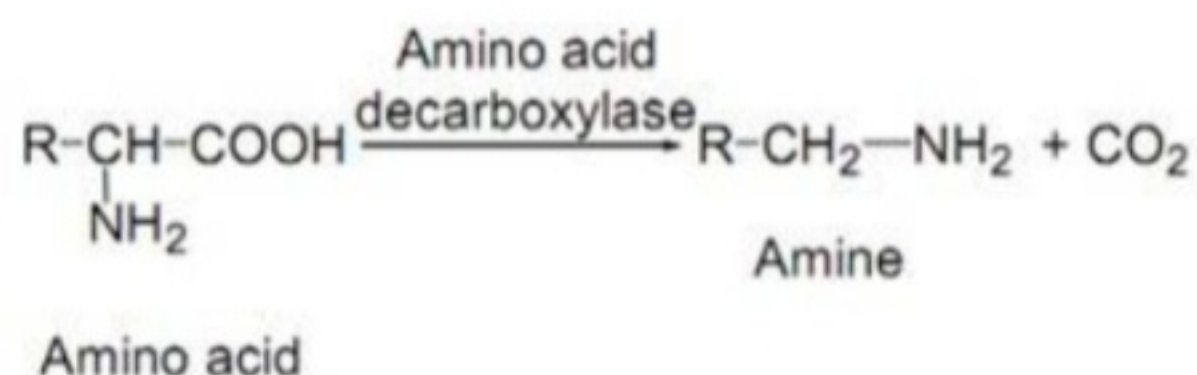


(c) Deamination of histidine: The enzyme histidase acts on histidine to liberate NH_3 by a non-oxidative deamination process.



3) DECARBOXYLATION:

This consider to the removal of CO_2 (carbon dioxide) from the carboxyl group of amino acids. The elimination of CO_2 requires the catalytic action of the pyridoxal phosphate coenzyme and enzymes decarboxylases. The enzymes work on amino acids resultant in the creation of the corresponding amines along with the liberation of CO_2 .



There are various amino acid decarboxylases make out in several tissues like liver, spleen, kidney, intestine, lung and brain. They translate the amino acids into the relevant amines and liberate CO_2 . For instance, histidine is transformed to histamine via the action of histidine decarboxylase.

The amino acid tryptophan is transformed to tryptamine, tyrosine to tyramine, etc. These types of amines are known biogenic amines that are physiologically significant.

